# Md. Taifur Hossain.

SOFTWARE ENGINEERING · COMPUTER VISION · ML

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Final-year Computer Science undergraduate who ships working software. Built and released **four open-source tools** across 3D vision, model compression, and developer tooling, each with tests, CI, documentation, and a tagged release, plus a **first-author paper accepted at ICCA 2026 (ACM)**. Also builds full-stack and applied-AI systems, from a Next.js learning platform to a hybrid medical expert system. Comfortable across C, C++, C#, Python, and JavaScript, and used to owning a problem from prototype through to a documented release. Open to software engineering internships across backend, full-stack, systems, and ML.

## TECHNICAL SKILLS

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|---|---|
| **LANGUAGES** | C, C++, C#, Python, JavaScript / TypeScript, SQL, Bash |
| **WEB & FRAMEWORKS** | Next.js, React, Node.js, Tailwind CSS, Flask, Streamlit, Three.js, Vite, Prisma, SQLite, HTML / CSS |
| **MACHINE LEARNING** | PyTorch, scikit-learn, NumPy, RAG and embeddings, Bayesian inference, heuristic search (A*), model quantization |
| **3D & VISION** | 3D Gaussian Splatting, NeRF, Nerfstudio, gsplat, COLMAP |
| **TOOLS & PRACTICE** | Git, Docker, GitHub Actions (CI/CD), GHCR, Linux, pytest, ruff, ESLint, Agile / Scrum |

## PROJECTS

### Lumina, AI Learning Management System

GitHub

*AI-assisted learning platform built across Agile sprints, with a retrieval tutor grounded in real course material.*

- Full-stack Next.js 16 and React 19 application in TypeScript, roughly **7,500 lines**, backed by Prisma and SQLite.
- Retrieval-augmented tutor that answers from uploaded course materials and shows the sources it used.
- Role-based access for Student, Instructor, and Admin, with JWT sessions, bcrypt password hashing, and Zod input validation.
- Adaptive learning paths, gamification (XP, badges, streaks), and instructor analytics dashboards.
- Runs fully offline against a mock AI provider, or against Google Gemini by setting a single environment variable.

**Tech Stack:** Next.js, React, TypeScript, Tailwind CSS, Prisma, SQLite, Google Gemini, Zod, Recharts.

### splat-slim

GitHub

*Training-free compression tool for trained 3D Gaussian Splatting models.*

- Compresses trained 3DGS models by **roughly 74% with under 3 dB quality loss**, with no retraining.
- Four-stage pipeline: adaptive opacity pruning, outlier clipping, spherical-harmonic reduction, and quantization.
- Three quantization modes: FP16, mixed FP16/INT8, and Morton-ordered sub-group INT8, the only one that writes per-group ranges to a sidecar `.meta.npz`. Unquantized output opens in standard viewers; quantized PLYs need a range-aware reader.
- Shipped as a CLI and a library, with automated tests and linting on every push and a tagged `v0.1.0` release.

**Tech Stack:** Python, Typer, NumPy, plyfile, pytest, ruff, GitHub Actions.

### AI-Based Medical Expert System

GitHub

*Hybrid expert system combining five AI techniques into a single diagnosis pipeline.*

- Combines rule-based reasoning with confidence factors, **A* search**, Best-First search, **Bayesian inference**, and a Decision Tree plus MLP neural network.
- Merges all five outputs through a weighted ensemble to produce a ranked diagnosis with confidence scores.
- Encodes 10 IF-THEN rules over 15 binary symptom features across 6 conditions, and benchmarks the pipeline against a real dataset.
- Serves the whole system through a Flask web interface with a symptom checklist and a results dashboard.

**Tech Stack:** Python, Flask, scikit-learn, NumPy.

### nerf-prep GitHub

*One-command Dockerized pipeline turning a folder of photos into a training-ready 3D dataset.*

- Packaged a fragile multi-tool chain (COLMAP to Nerfstudio) into **a single reproducible Docker command**, published to GHCR automatically by CI.
- Automatic matcher selection by capture type, plus GPU and CPU auto-detection with a fallback, so the same command runs on a cloud GPU or a laptop.
- Validates input before a long run, flagging double extensions and too-few images, then prints a registration-rate summary. Verified on a real 40-photo capture at **40 of 40 registered**.

**Tech Stack:** Python, Typer, Docker, COLMAP, Nerfstudio, GitHub Actions, GHCR.

### paper-chat GitHub

*Retrieval-augmented question answering over a PDF library, with citations you can verify.*

- **Every answer cites the exact paper and page** it came from, and the app declines to answer when the sources do not cover the question.
- Parses PDFs per page so each retrieved chunk carries a (paper, page) tag, enabling cross-paper comparison questions.
- Swappable LLM backend (Groq or Gemini) behind a small interface, with retrieval and citation logic unit-tested using fakes, so tests run with no API key.

**Tech Stack:** Python, Streamlit, sentence-transformers, PyMuPDF, NumPy.

### splat-lab Live Demo · GitHub

*Browser-based, real-time demo of the compression pipeline.*

- Load a 3D scene and watch file size and visual quality trade off **live on the GPU** while dragging sliders.
- Ported the Python compression maths to TypeScript so the browser reproduces the same results, running entirely client-side with no backend.

**Tech Stack:** TypeScript, Three.js, Vite, GitHub Pages.

## OPEN SOURCE CONTRIBUTIONS

- **gsplat PR #1063** (open): opacity × area weighted K-means for the `PngCompression` SH codebook, **+0.111 dB** mean PSNR on 9 MipNeRF360 scenes and +0.052 dB on Tanks & Temples, better on every scene at the same size and format.
- **gsplat PR #1061** (open): fixed `PngCompression` failing on `sh_degree=0` scenes and empty tensors.

## PUBLICATION

**Beyond Naive INT8: Adaptive Post-Training Compression for 3D Gaussian Splatting** · ACCEPTED

M. T. Hossain, S. S. Sourav, M. A. Hossain. Proc. ICCA 2026 (ACM), Dhaka.

- First author. Built the full experimental pipeline and evaluated it across three scenes on a single 16 GB GPU.
- Achieved a **74% size reduction without retraining**, and diagnosed a quantization failure mode that a sub-group range scheme recovers to within 0.1 to 0.7 dB.

## EDUCATION

**International University of Business Agriculture & Technology (IUBAT)** — Dhaka, BD

B.Sc. in Computer Science & Engineering (BCSE) · CGPA 3.3 / 4.00 — Expected 2027

**Higher Secondary Certificate (HSC)** · GPA 5.00 / 5.00 — Dhaka, BD · 2019 - 2021

**Secondary School Certificate (SSC)** · GPA 5.00 / 5.00 — Dhaka, BD · 2018

## LANGUAGES

| | |
|---|---|
| ENGLISH | Proficient |
| BANGLA | Native |

## REFERENCES

**Dr. Ishtiak Al Mamoon** — Professor

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